PROJECT CARESYNC

ELECTRICAL AND COMPUTER ENGINEERING
UNIVERSITY OF AVEIRO

CareSync

Integrated Medication Management Ecosystem

Safer medication. Greater independence.
A more connected tomorrow.



CareBox
Smart Medication Dispenser

CareBand
Wearable Interaction Device

CareApp
Mobile Application

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The Problem

CareSync

Medication management still relies on human memory and routine.

MISSED DOSES
Timing & Forgotten intake



Medication may be forgotten or taken outside the prescribed schedule.

DOSAGE MISMANAGEMENT
Critical Dosing Errors



Incorrect medication, dosage or timing can compromise treatment safety.

POLYPHARMACY COMPLEXITY
Overwhelming Schedules



Managing multiple medications across different daily schedules can be challenging.

INFORMATION VOID
Zero Real-Time Visibility



Limited real-time visibility makes medication routines difficult to coordinate across users and care teams.

The real challenge is not simply reminding someone to take their medication. It is creating a clear, more connected view of the medication process.

The idea - CareSync Ecosystem

An integrated medication management ecosystem designed to support people who may have difficulties managing medication independently.

Designed for people who may experience:



COGNITIVE CHALLENGES
e.g. memory loss or Alzheimer's disease

REDUCED INDEPENDENCE
e.g. older adults living alone

SENSORY OR PHYSICAL LIMITATIONS
affecting interaction with medication



CAREBOX — Automated Dispensing
Physical hardware unit for scheduled dosage control and local feedback.



CareSync



CAREBAND — Wearable Interaction
Low-friction wearable for patient identification and intake confirmation.

CAREAPP — Digital Platform
Centralized software for SNS prescription parsing, schedules, and remote tracking.

One ecosystem. Three connected components.

CareBox - Smart Medication Dispenser

CareSync

Physical hardware unit engineered for automated scheduled dispensing, RFID verification, and local user feedback.

1 — AUTOMATED DISPENSING:
Dispenses scheduled medication according to the configured medication plan.

2 — RFID IDENTIFICATION:
RFID-based authentication for precise user identification and medication-event verification.

3 — MULTI-SENSORY FEEDBACK:
Delivers clear acoustic and visual alerts during active medication windows.

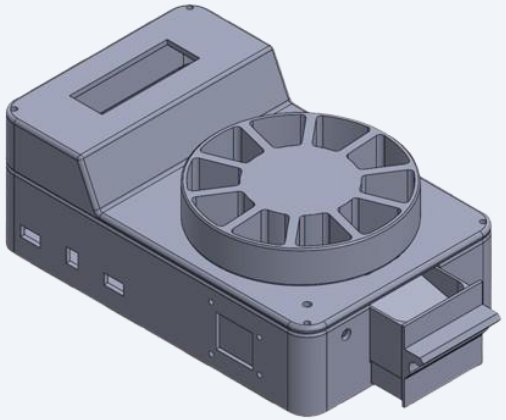
4 — STATUS & WIRELESS TELEMETRY:
Detects device status and syncs real-time events with the CareSync ecosystem.

5 — ACCESSIBLE INTERACTION:
A proposed touchscreen interface with intuitive icons and audio guidance to support users with different cognitive, sensory and physical needs.

6 — MODULAR REFILL SYSTEM:
Future replaceable, pre-filled and sealed medication modules designed to simplify replenishment and support institutional use.



Physical 3D-Printed Prototype



3D CAD Model (SolidWorks)

CareBand – Wearable Interface

Accessible, low-friction wearable designed for user identification, intake confirmation, and feedback.

CareSync

1 — USER IDENTIFICATION & AUTHENTICATION:

Identifies and authenticates the user during the medication process to prevent errors.

2 — INTAKE CONFIRMATION & FEEDBACK:

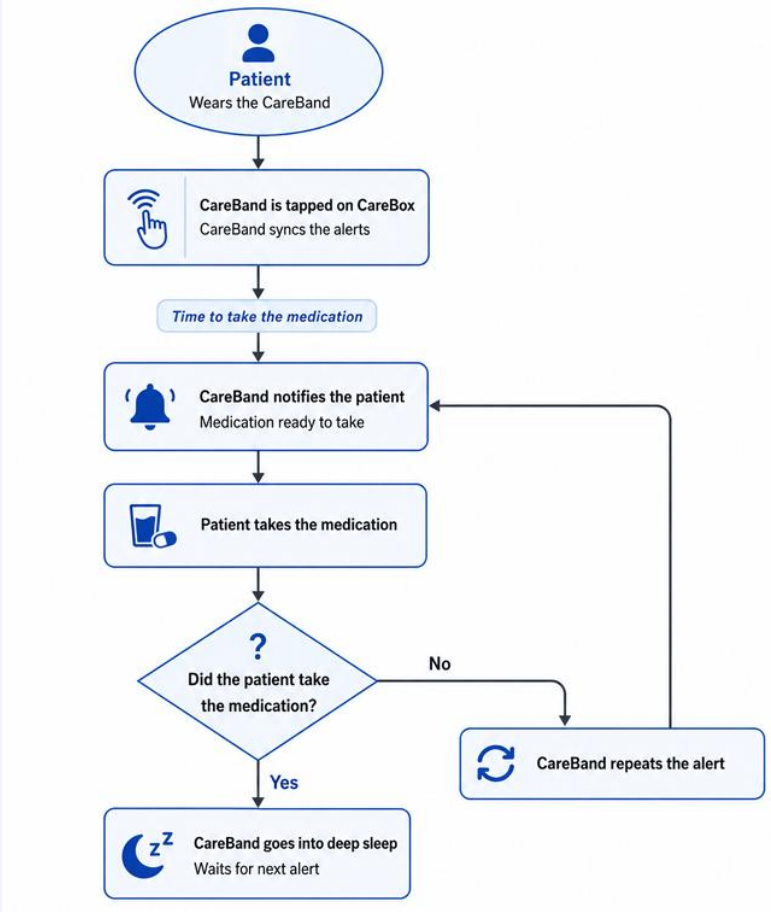
Enables immediate intake confirmation with direct visual/haptic notifications when interaction is required.

3 — ACCESSIBLE & LOW-COGNITIVE DESIGN:

Minimizes physical and cognitive effort for elderly users or patients with physical/sensory limits.

4 — WIRELESS CAREBOX INTEGRATION:

Seamlessly exchanges real-time medication events and telemetry with the **CareBox** unit.



INTERACTION FLOW:

Medication Ready Alert → User Identification → Physical Confirmation → Event Logged

CareApp – Core Capabilities & Prescription to plan

CareSync

From a prescription image to a structured medication plan.
End-to-end software platform connecting prescription parsing, medication tracking, and remote monitoring.

1 — PRESCRIPTION PARSING & SCHEDULE:

Captures SNS prescriptions and converts data into a structured medication schedule.

2 — REAL-TIME MONITORING & ALERTS:

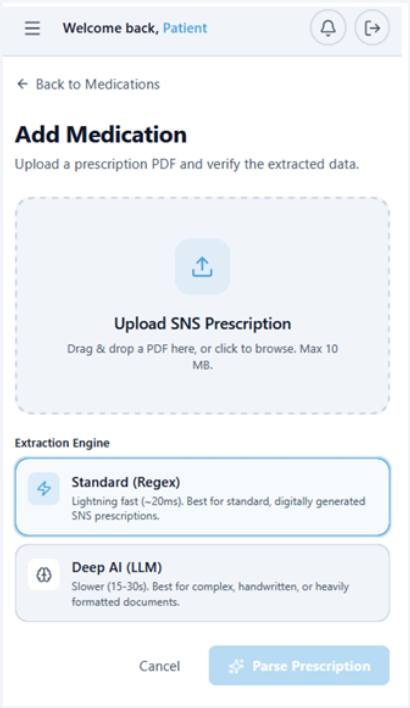
Provides live status of scheduled doses and sends instant notifications for missed or unconfirmed events.

3 — ADHERENCE & HISTORY TRACKING:

Logs timestamped intake events to generate adherence trends and history report telemetry.

4 — REMOTE CARE DASHBOARD:

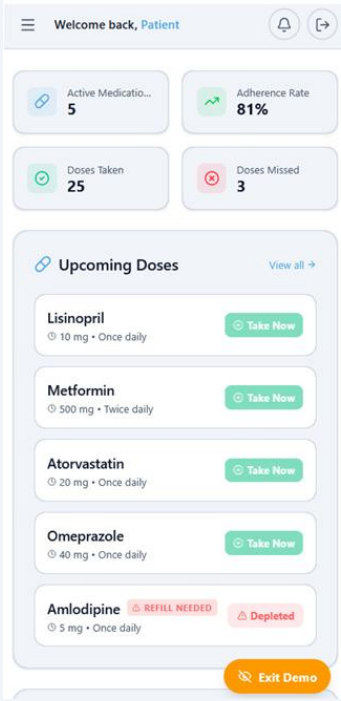
Centralized interface with role-based access for caregivers and healthcare professionals to monitor multiple users.



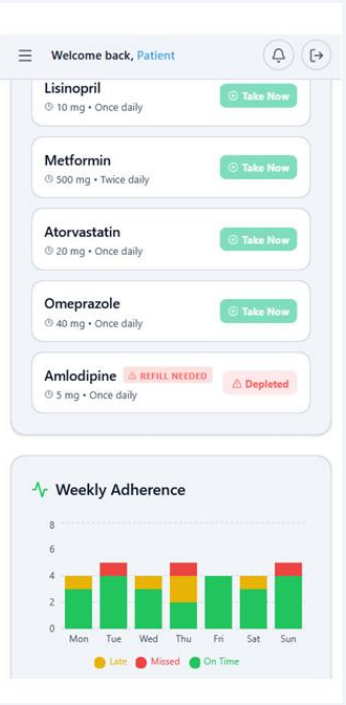
1 - SNS Prescription Upload



2 - Automated Calendar Sync



3 - Adherence & Telemetry



4 - Remote Monitoring Dashboard

INNOVATION & PRIOR-ART RESEARCH:

The prescription extraction methodology was investigated through a formal prior-art search using Google Patents, including benchmarking against existing solutions.
Goal: assess novelty, differentiation and potential intellectual property protection.

The CareSync Ecosystem

Three connected components working as one medication management system.

CAREAPP



Manage & Monitor

- Medication schedules
- Patient information
- Medication history
- Notifications

Wi-Fi / Internet

CAREBOX



Dispense & Control

- Automated dispensing
- Medication events
- User identification
- Local feedback

Bluetooth Low Energy

CAREBAND



Interact & Confirm

- User interaction
- Medication confirmation
- User identification
- User notifications

FROM INFORMATION TO ACTION



→ Medication Schedule
→ CareBox Dispensing
→ User Interaction
→ Medication Event
→ CareApp Monitoring

Each component has a specific role, while communication between them enables a connected medication management process.

CareSync

End-to-End User Flow

From prescription parsing to physical dispensing and telemetry monitoring.

INPUT & SCHEDULING:



01- Prescription:
Prescription information is captured and processed to create the medication schedule.

02- Schedule:
Medication, dosage and administration times are organized in the CareApp.

PREPARATION & DISPENSING:



03- CareBox Prep: Dose organization based on schedule.

04- Medication Time: Dose availability with local feedback (acoustic & visual prompts signal dosage window.).

USER INTERACTION:



05- Interaction: User engages with **CareBand** during process.

06- Event Tracking: Real-time intake status logging (taken, missed, etc.); Dispenser confirms compartment opening and dosage intake.

MONITORING & TELEMETRY:



07- System Monitoring: Real-time sync to **CareApp** for full visibility.

08- Caregiver / Clinician Dashboard: Full visibility and immediate alert telemetry for health professionals.

CLOSED-LOOP MEDICATION MANAGEMENT:

Prescription → Medication Schedule → Refill Preparation → Dispensing → User Interaction → Intake Verification → Real-Time Event Sync → Caregiver/Clinician Dashboard

A day with CareSync

CareSync

CARESYNC IN PRACTICE - A day in a residential care home:

1. MEDICATION PLANNING

Physicians prescribe treatments, and prescriptions are processed via CareApp to configure each resident's medication schedule.

2. PERSONALIZED REFILL PREPARATION

Pre-filled, individually identified and sealed medication modules are prepared for each resident, with appropriate protection against moisture and mix-ups.

3. ACCESSIBLE MEDICATION DISPENSING

At the scheduled time, CareBox guides the resident through the medication process using a simple touchscreen interface, intuitive icons and audio instructions.

4. MEDICATION-EVENT VERIFICATION

CareBox records dispensing and user-interaction events. Additional camera-assisted verification could provide further evidence of the medication-taking process.

5. REAL-TIME CARE MONITORING

Authorized nurses, caregivers and healthcare professionals can access updated medication statuses, review events and receive alerts when a dose requires attention.

ONE ECOSYSTEM. COORDINATED CARE.

Potential impact: More organized medication routines, improved visibility for care teams and a scalable refill-based service model for residential care facilities.

Target Users / Potential Customers

CareSync

Designed around people who may need additional support with medication management.



TARGET USERS:

Older Adults: Maintaining independence with daily medication support.

Cognitive Challenges: Assisting patients with memory loss or cognitive impairment.

Sensory / Physical Limits: Accessible, guided multi-sensory interaction.

Complex Routines: Managing multi-prescription schedules seamlessly.



POTENTIAL CUSTOMERS:

Families & Caregivers: Real-time visibility over routines and intake events.

Home-Care Providers: Seamless integration into assisted-living services.

Pharmacies & Networks: Value-added medication management platform.

Insurers & Health Systems: Digital solution to reduce non-adherence costs.



VALUE PROPOSITION

Users: Greater independence & safer routines.
Caregivers: Real-time visibility & peace of mind.
Organizations: Connected adherence platform & telemetry.

Future Potential

CareSync

From an academic prototype to a scalable connected-care solution.



PHASE 1:
SHORT-TERM Strengthen the Core

- Redesigning CareBox for enhanced modularity and ease of use.
- Touch interface with intuitive icons and audio guidance.
- Higher reliability in dispensing and wireless communication.
- Improved medication intake logging and status tracking.



PHASE 2:
MEDIUM-TERM Connect & Scale

- Real-time monitoring dashboards for caregivers and authorized clinicians.
- Remote alerts and companion tracking.
- Preparation and distribution workflow for refill modules.
- Multi-user and institutional support.



PHASE 3:
LONG-TERM Intelligence & Integration

- Privacy-by-design camera-assisted intake verification.
- Predictive medication adherence analytics.
- Integration with pharmacies and healthcare systems.
- Expansion of the business model to healthcare organizations.



LONG-TERM VISION: A connected medication management ecosystem combining physical automation, wearable interaction, and intelligent digital monitoring — expanding into broader home-based healthcare workflows.

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CONNECTED MEDICATION MANAGEMENT

Thank You!

Thank you for your time and interest in our solution and in this presentation.

If you have any questions or would like to know more, please feel free to contact me.

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BETTER CARE. TOGETHER.



CareSync
Together for
better health.